

Žan Grad – Curriculum Vitae

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BASIC INFORMATION

- Nationality: Slovenian
- Place and date of birth: Celje (Slovenia), October 6 1993
- Current institution: Instituto Superior Técnico, University of Lisbon

EDUCATION

Doctoral degree in Mathematics (Ph. D.) 2021 – ongoing

Instituto Superior Técnico, University of Lisbon

- Expected defense: June 2025.
- Member of Centre for Mathematical Analysis, Geometry, and Dynamical Systems.
- Topic: *Fundamentals of Lie categories and Yang–Mills theory for multiplicative Ehresmann connections.*
- Supervisors: Ioan Mărcuț and Pedro Resende.

Master of Mathematics (M. Sc.) Graduated in 2020

University of Ljubljana, Faculty of Mathematics and Physics

- Master's thesis: *Gauge field theory, Yang–Mills–Higgs equations and spin structures on Lorentzian manifolds.*
- Thesis awarded with the Prešeren award of University of Ljubljana.
- Supervisor: Sašo Strle.

Bachelor of Physics (B. Sc.) Graduated in 2017

University of Ljubljana, Faculty of Mathematics and Physics

- Bachelor's thesis: *Wave function collapse upon measuring the number of photons in a cavity.*
- Supervisor: Tomaž Rejec.

TEACHING EXPERIENCE

Assistant professor of mathematics Sep 2020 – Sep 2021

University of Ljubljana, Faculty of Mathematics and Physics

- Employed full time for two semesters, with an average load of 10 hours of teaching per week.
- Held exercise sessions, constructed written exams, graded students.
- Taught two master-level courses: Analysis on manifolds and Differential geometry.
- Taught five bachelor-level courses: Analysis IV for financial mathematicians, Mathematics 2 for physicists, Mathematics 4 for physicists, Basics of mathematical analysis for computer engineers, Mathematics for pharmacists.

Recorded exercise sessions for these courses are available on my personal webpage, in the [teaching section](#).

PUBLICATIONS

- [1] Žan Grad, *Fundamentals of Lie categories*, J. Noncommut. Geom. **19** (2025), no. 1, 211–248, DOI 10.4171/JNCG/563. ([arXiv](#)).
- [2] ———, *Covariant derivatives in the representation-valued Bott–Shulman–Stasheff and Weil complex*. ([arXiv](#)).
- [3] ———, *Yang–Mills theory for multiplicative Ehresmann connections*. In preparation, to appear in Spring 2025.

TALKS

- Planned: *Double complexes of representation-valued forms*, University of Illinois Urbana–Champaign ([Symplectic and Poisson Geometry Seminar](#)), April 28, 2025.
- *Double complexes of representation-valued forms*, Universität zu Köln, [Poisson Geometry Seminar](#), February 5, 2025.
- *Double complexes of representation-valued forms*, JMU Würzburg, [Oberseminar Deformationsquantisierung und Geometrie](#), January 31, 2025.
- *The exterior covariant derivative of multiplicative forms*, IMPA, Rio de Janeiro, November 14, 2024.
- *Crash course on Symplectic fibrations*, Radboud University Nijmegen, April 2024.
- *Mini course on Lie categories*, [Lie groupoid and algebroid week in Coimbra II](#), July 2023.
- *Lie categories*, [Geometry Seminar](#) of Lisbon, Instituto Superior Técnico, Universidade de Lisboa, May 9, 2023.
- *Lie categories*, PhD seminar of Radboud University Nijmegen, April 2023.
- Two talks on *Lie categories*, [Topology Seminar](#) of Faculty of Mathematics and Physics, University of Ljubljana, March 20 and March 27, 2023.
- A series of lectures on Lie groupoids, Instituto Superior Técnico, Universidade de Lisboa, Spring semester 2021/22.
- *Mini course on Minkowski space and the theory of relativity*, Summer school of Mathematics and Physics at University of Ljubljana, August 2019.

In the spring semester of 2024, I organized an online seminar on the integration of Lie algebroids to Lie groupoids, where we discussed and proved the well-known result by Crainic and Fernandes stating the precise obstructions to the integrability of Lie algebroids. This seminar was co-organized by fellow PhD students from University of Coimbra and Radboud University.

CONFERENCES AND WORKSHOPS

The following list contains attended and planned conferences and workshops.

- Planned: [Interactions of Poisson Geometry, Lie Theory and Symmetry](#), June 30–July 4, 2025 (Lisbon).
- [Higher Geometric Structures along the Lower Rhine XVIII](#), January 23–24, 2025 (Nijmegen).
- [XI Workshop on Poisson Geometry and Related Topics](#), November 4–8, 2024 (Mannheim).
- [XXXII International Fall Workshop on Geometry and Physics](#), September 2–5, 2024 (Coimbra).
- [Poisson School & Conference 2024](#), July 1–12, 2024 (Naples).
- [Lie groupoid and algebroid week III](#), June 3–7, 2024 (Coimbra).
- [Higher Geometric Structures along the Lower Rhine XVII](#), May 2–3, 2024 (Bonn).
- [Lie groupoid and algebroid week II](#), July 10–14, 2023 (Coimbra).
- [The many interactions of symplectic and Poisson geometry](#), June 19–23, 2023 (Paris).
- [Lie groupoid and algebroid week I](#), September 26–30, 2022 (Coimbra).
- [Poisson School & Conference 2022](#), July 18–29, 2022 (attended online).

POSTERS

- [The exterior covariant derivative of multiplicative forms](#) for XI Workshop on Poisson Geometry and Related Topics (Poisson in the Amazon), November 2024.

GRANTS

- [COST CaLISTA](#) European grant for short-term scientific missions. This grant enabled me to visit Ioan Mărcuț in person in Nijmegen, in the period of April and May 2024.
- PhD grant [UI/BD/152069/2021](#) by Fundação para a Ciência e a Tecnologia (FCT), Portugal.

AWARDS

- *Prešeren award of Faculty of Mathematics and Physics, University of Ljubljana*, for master's thesis *Gauge field theory, Yang–Mills–Higgs equations and spin structures on Lorentzian manifolds*, 2020.

MENTORING

- Master's student: *Matthijs Lau*, Radboud University Nijmegen. Topic: *Symplectic groupoid fibrations*. Expected to graduate in 2025. This is a joint supervision with Ioan Mărcuț.

OTHER

Physics

I am interested in applying modern mathematical methods to describe the theory of general relativity, quantum mechanics, many-body quantum mechanics, quantum field theory, classical mechanics, classical field theory (hydrodynamics, electromagnetism, elastomechanics and gauge field theory).

Languages

Native speaker of Slovenian language. I am fluent in both written and spoken English.

Programming languages

I possess advanced knowledge of languages of the web: PHP, Javascript, HTML, CSS. Principles of backend and frontend frameworks, together with concrete advanced implementations of frameworks Laravel (PHP) in VueJS (Javascript). To support myself, I worked a part-time job as a full-stack web developer between 2010 and 2016. I also have basic knowledge of Mathematica and Python and some familiarity with C.